

PHASE23_CLOSURE.md

DSN Bootcamp Qualification Hackathon 2026 ML Track

Phase 23 Closure Report

Report Date: 2026-09-22

Phase 23 Objective

Conduct consensus robustness search for the canonical submission, evaluating feature combinations across multiple validation protocols and seeds.

Methodology

- **Protocols tested:** SKF, PMH, TSM
- **Seeds tested:** 42, 123, 999
- **Feature combinations:** 144 rows, 16 feature sets
- **Dominance audit:** 9-cell incremental deltas (180 rows)

Key Findings

1. **Canonical control reproduced exactly** for all 3 seeds across all protocols.
2. **No globally dominant combination** found across all 9 validation cells.
3. **E+G is NOT globally dominant** (1/9 cells worsened).
4. **Simplest evidence-supported candidate:** E_alone (1 feature, 7/9 improved).
5. **All Stage 3 deployment safety checks passed** (18/18).

Stage 3 Test Evaluation

- CONTROL, E_alone, and E+G predictions generated (1705 rows each)
- 100% unseen product-store pairs in test set
- Product-store multiplicity drops 4.38 → 1.58
- Single-store product fraction 2.96% → 56.38%
- Prediction shift audit: E+G vs E_alone mean shift=+3.49, std=93.54

Integrity Verification

- Script consistency: PASS
- Artifact consistency: PASS
- Raw prediction preservation: PASS
- Canonical immutability: PASS
- 0 contradictory statements

Leaderboard Submission

- Authorized E+G test submission attempted
- **BLOCKED:** Kaggle CLI, Python package, and API credentials all unavailable

Closure Status

Phase 23 is **CLOSED**. No further experiments authorized. Canonical submission remains frozen. E3 retained as alternate.

Report generated as part of FINAL COMPETITION RELEASE / REPRODUCIBILITY WAR ROOM.